

# Microlocal Morse Theory and Truncated Microsupport

Toshi2019

December 2, 2025\*

## Preface

In these notes, We study microlocal Morse theory, which is developed by Guillermou, Kashiwara, and Schapira in [GKS12], but replace microsupports with truncated microsupprts introduced by Kashiwara, Monteiro-Fernandes, and Schapira in [KMS03].

## 1 Truncated Microsupport

**Definition 1.1.** (i) Let  $F \in D^b(\mathbf{k}_X)$ . The closed conic subset  $SS_k(F)$  of  $T^*X$  is defined by:  $p \notin SS_k(F)$  if and only if  $F$  satisfies the equivalent conditions in Prop.

(ii) aa

## 2 Results

**Proposition 2.1.** Let  $F$  in  $D^b(\mathbf{k}_M)$ , and  $\psi: M \rightarrow \mathbf{R}$  be a real-valued function of class  $C^1$ , proper on  $\text{supp}(F)$ . Let  $a < b$  be real numbers. We assume  $d\psi(x) \notin SS_k(F)$  for  $a \leq \psi(x) < b$ . For  $t \in \mathbf{R}$ , we set  $M_t := \psi^{-1}(]-\infty, t])$ . Then, the restriction morphisms

$$H^j(M_b; F) \rightarrow H^j(M_a; F)$$

are isomorphic for each  $j \leq k$ .

**Proof.** First, we have

$$H^j(M_t; F) \cong H^j \text{R}\Gamma(]-\infty, t[; \text{R}\psi_* F)$$

for  $t \in I$ . Since we assume that  $d\psi(x) \notin SS_k(F)$  for  $a \leq \psi(x) < b$ , and by the formula  $SS_k(\text{R}\psi_* F) \subset \psi_\pi \psi_d^{-1} SS_k(F)$ , we have

$$(H^j \text{R}\Gamma_{[a, b[} \text{R}\psi_* F)_{\psi(x)} = 0$$

---

\*2025/03/09 first draft

for each  $j \leq k$ . Therefore, letting  $d \in \mathbf{Z}$  be the minimum number that

$$H^d \mathrm{R}\Gamma_{]-\infty, a[}(\mathrm{R}\psi_* F) \neq 0,$$

by the long exact sequence

$$\begin{aligned} 0 &\rightarrow H^d \mathrm{R}\Gamma_{[a, b[}(\mathrm{R}\psi_* F) \rightarrow H^d \mathrm{R}\Gamma_{]-\infty, b[}(\mathrm{R}\psi_* F) \rightarrow H^d \mathrm{R}\Gamma_{]-\infty, a[}(\mathrm{R}\psi_* F) \\ &\rightarrow H^{d+1} \mathrm{R}\Gamma_{[a, b[}(\mathrm{R}\psi_* F) \rightarrow H^{d+1} \mathrm{R}\Gamma_{]-\infty, b[}(\mathrm{R}\psi_* F) \rightarrow H^{d+1} \mathrm{R}\Gamma_{]-\infty, a[}(\mathrm{R}\psi_* F) \\ &\rightarrow \dots \\ &\rightarrow H^k \mathrm{R}\Gamma_{[a, b[}(\mathrm{R}\psi_* F) \rightarrow H^k \mathrm{R}\Gamma_{]-\infty, b[}(\mathrm{R}\psi_* F) \rightarrow H^k \mathrm{R}\Gamma_{]-\infty, a[}(\mathrm{R}\psi_* F) \\ &\rightarrow H^{k+1} \mathrm{R}\Gamma_{[a, b[}(\mathrm{R}\psi_* F) \rightarrow \dots, \end{aligned}$$

we have isomorphisms

$$(H^j \mathrm{R}\Gamma_{]-\infty, b[}(\mathrm{R}\psi_* F))_{\psi(x)} \xrightarrow{\sim} (H^j \mathrm{R}\Gamma_{]-\infty, a[}(\mathrm{R}\psi_* F))_{\psi(x)}$$

for any  $j \leq k$  and  $a \leq \psi(x) < b$ , and then,

$$H^j \mathrm{R}\Gamma_{]-\infty, b[}(\mathrm{R}\psi_* F) \xrightarrow{\sim} H^j \mathrm{R}\Gamma_{]-\infty, a[}(\mathrm{R}\psi_* F) \quad (2.1)$$

for all  $j \leq k$ . Then we can conclude that

$$H^j(M_b; F) \xrightarrow{\sim} H^j(M_a; F)$$

by applying to the equality (2.1) the functor  $\Gamma(]-\infty, b[; \cdot)$  and calculating as follows:

$$\begin{aligned} \Gamma(]-\infty, b[; H^j \mathrm{R}\Gamma_{]-\infty, b[}(\mathrm{R}\psi_* F)) &\cong \Gamma_{]-\infty, b[}(\Gamma(]-\infty, b[; \mathrm{R}^j \psi_* F)) \\ &\cong \Gamma(]-\infty, b[; \mathrm{R}^j \psi_* F) \\ &\cong H^j(\psi^{-1}(]-\infty, b[); F) \\ &\cong H^j(M_b; F) \end{aligned}$$

and

$$\begin{aligned} \Gamma(]-\infty, b[; H^j \mathrm{R}\Gamma_{]-\infty, a[}(\mathrm{R}\psi_* F)) &\cong \Gamma_{]-\infty, a[}(\Gamma(]-\infty, b[; \mathrm{R}^j \psi_* F)) \\ &\cong \Gamma(]-\infty, a[; \mathrm{R}^j \psi_* F) \\ &\cong H^j(\psi^{-1}(]-\infty, a[); F) \\ &\cong H^j(M_a; F). \end{aligned}$$

□

### 3 Truncated Morse Inequalities

The main theorem of [ST92] is the following: we assume

$$\begin{aligned} &\psi^{-1}(]-\infty, t[) \cap \mathrm{supp}(F) \text{ is compact for any } t \in \mathbf{R}, \\ &\text{there is a finite family } \{K_j\}_{1 \leq j \leq N} \text{ of disjoint compact} \end{aligned} \quad (\text{H.1})$$

subsets in  $X$  and real numbers  $\{s_j\}_{1 \leq j \leq N}$  such that

$$\pi(\Lambda_\psi \cap \Lambda) = \bigcup_j K_j, \quad \psi|_{K_j} = s_j, \quad (\text{H.2})$$

$$V_j := \mathrm{R}\Gamma(K_j; \mathrm{R}\Gamma_{\{\psi \geq s_j\}}(F)) \text{ belongs to } \mathrm{D}^b(\mathrm{Mod}^f(\mathbf{k})). \quad (\text{H.3})$$

Set

$$n_l := \sum_j \dim H^l(V_j). \quad (3.1)$$

Then

**Theorem 3.1** (Generalized Morse Inequalities [ST92, Theorem 1.1]). Assume (H.1), (H.2), and (H.3). Then

- (i)  $R\Gamma(X; F) \in D^b(\text{Mod}^f(\mathbf{k}))$ ,
- (ii) setting  $b_l(X, F) = b_l(R\Gamma(X; F))$ , we have the inequalities

$$b_l^*(X, F) \leq n_l^*$$

where  $n_l$  is given by (3.1).

## References

- [GKS12] Stéphane Guillermou, Masaki Kashiwara, Pierre Schapira, *Sheaf Quantization of Hamiltonian Isotopies and Applications to Nondisplaceability Problems*, Duke. Math. J. 161, No. 2, pp.201–245, 2012.
- [KMS03] Masaki Kashiwara, Teresa Monteiro Fernandes, Pierre Schapira, *Truncated Microsupport and Holomorphic Solutions of D-modules*, Ann. Scient. Éc. Norm. Sup., série 4, tome 36, pp.583–399, 2003.
- [KS90] Masaki Kashiwara, Pierre Schapira, *Sheaves on Manifolds*, Grundlehren der Mathematischen Wissenschaften, 292, Springer, 1990.
- [ST92] Pierre Schapira, Nobuyuki Tose, *Morse Inequalities for  $\mathbf{R}$ -constructible Sheaves*, Adv. Math. 93, pp.1–8, 1992.